

AIRWISE

India Air Quality Intelligence Dashboard

Professional Project Report

Framework	Streamlit
Data Processing	Pandas, NumPy
Visualization	Plotly Express, Matplotlib
Dataset	Air_quality_in_india.csv (3,514 records)
Report date	September 2026

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All figures in this report are computed directly from the supplied dataset and application code.

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1. Introduction

AIRWISE — India Air Quality Intelligence Dashboard is a Streamlit-based analytics application built to make air-quality observations easy to explore at the national, state, city, station and pollutant level. It combines data cleaning, descriptive analysis, interactive visualization, geographic exploration and automatically generated data-driven insights in a single, self-contained app.

This revision of the report documents both the original design and a set of fixes and interface improvements applied after review: a data-loading bug, a broken chart caused by a data-shape assumption, a filter-selection logic bug, and a full redesign of the navigation and filter experience.

2. Objectives

- Inspect, clean and validate the supplied air-quality dataset.
- Provide a clear, KPI-driven overview of pollutant levels across India.
- Enable geographic exploration using station latitude/longitude data.
- Compare pollutant levels across states, cities and monitoring stations.
- Generate insights directly from the observed dataset, with no fabricated forecasts.
- Present the analysis through a robust, attractive, and genuinely interactive interface.

3. Dataset Overview

The application reads **Air_quality_in_india.csv**. Key structural facts, computed directly from the file:

Metric	Observed Value
Raw rows	3,514
Raw columns	11
States represented	30
Cities represented	261
Monitoring stations	502
Pollutant types	7 (CO, NH3, NO2, OZONE, PM10, PM2.5, SO2)
Timestamp coverage	22-09-2026 22:00:00 (single snapshot — see Section 5)
Duplicate rows	0

Core fields: *country, state, city, station, last_update, latitude, longitude, pollutant_id, pollutant_min, pollutant_max, pollutant_avg*.

4. Data Cleaning & Quality Checks

4.1 Missing values

The raw file contains 990 missing cells, concentrated entirely in the three pollutant measure columns. The application imputes these using the column median, computed after filtering.

Column	Missing cells	Treatment
pollutant_min	330	Median imputation
pollutant_max	330	Median imputation
pollutant_avg	330	Median imputation

4.2 Other checks

- Duplicate rows: 0 fully duplicated rows found; the app does not currently drop duplicates as a result, since none were present to remove.
- Timestamp parsing: *last_update* is parsed as DD-MM-YYYY HH:MM:SS with invalid values coerced to missing and dropped; 0 rows failed to parse.
- Text fields (state, city, station, pollutant_id) are stripped of stray leading/trailing whitespace as part of the current cleaning step, preventing duplicate categories caused by formatting alone.

5. What Was Fixed in This Revision

A review of the running app surfaced three concrete defects. Each is documented below with the root cause and the applied fix.

5.1 — Blank “Over Time” chart

Before: The National Overview line chart appeared empty, with the x-axis showing sub-second timestamps (e.g. 23:59:59.9995) instead of a readable date range.

After: Root cause: every row in the dataset carries the identical timestamp (22-09-2026 22:00:00) — this is a single real-time snapshot, not a time series, so a single-point line chart has nothing to draw and Plotly auto-zoomed to microseconds. The app now detects a single-timestamp dataset and replaces the broken trend chart with a “Top 15 stations” ranking bar chart and a category breakdown donut chart, and shows an explanatory notice to the user.

5.2 — Data file not found

Before: The loader looked for the exact filename 'Air quality in india.csv' (with spaces). The actual uploaded file was named Air_quality_in_india.csv (underscores), which would fail to load wherever the two names don't match exactly.

After: The loader now checks several known name variants, falls back to a normalized filename search across the working directory, and finally offers a file-uploader widget if no match is found — so the app never hard-crashes on a naming mismatch.

5.3 — Incorrect “most prevalent pollutant” logic

Before: The KPI card used `len(selected_pollutant) == 1` to decide whether to show the most common pollutant. Since `selected_pollutant` is a plain string (from a single-select dropdown), this checked the string's character length, not the number of pollutants selected — so the intended branch never actually executed.

After: Replaced with a direct KPI card that always shows the currently selected pollutant, and moved the “most prevalent pollutant” statistic into the Overview KPI row where it is computed correctly with `.mode()`.

6. Dashboard Features

Tab	Purpose
Overview	KPI summary cards, top-15 station ranking for the selected pollutant, and a category breakdown donut chart.
Map	Geographic scatter map of stations, colour-switchable between raw concentration and AQI-style category.
Pollutant Analysis	Grouped bar chart comparing pollutants by state, and a box plot of each pollutant's distribution.
State & City Comparison	Top/bottom-10 city rankings, full state ranking, an inline city search box, and a filtered-data CSV download.
Insights	Automatically generated, deterministic textual insights computed from the currently filtered data — not a machine-learning forecast.

All pollutant readings are additionally classified into a CPCB-style category (Good, Satisfactory, Moderate, Poor, Very Poor, Severe) for PM2.5 and PM10, where the standard breakpoints are well-established for the dataset's microgram-per-cubic-metre units. Other pollutants are labelled with a dataset-relative Low/Moderate/High/Very High scale, since their exact reporting units are not confirmed in the source file.

7. Interface & Filter Redesign

7.1 Navigation

The dashboard was restructured from five stacked, scrolling sections into five tabs (Overview, Map, Pollutant Analysis, Comparison, Insights), reducing scroll depth and letting a user jump directly to the view they need.

7.2 Filter system

The original sidebar rendered every one of the 30 states and every one of the 261 cities as individual tags by default, which produced a cluttered, hard-to-scan filter panel (visible in the original screenshot review). The filter system was redesigned around an explicit scope choice:

- **All India** (default) — no selections required, nothing to scroll through.
- **Specific state(s)** — a searchable multiselect appears only when chosen; matching cities are derived automatically.
- **Specific city(ies)** — the inverse path, for users who think in cities rather than states.

7.3 Visual design

- A consistent navy/teal colour system (matching the CPCB-style category palette) is used across KPI cards, headers, charts and the map.

- Text-based icons and emoji were removed from headings, tabs, buttons and messages for a more formal, professional look.
- KPI values are shown as styled cards rather than plain Streamlit metrics, for stronger visual hierarchy.

8. Methodology

The workflow follows a practical exploratory-data-analysis pipeline: load the CSV (with a resilient file-matching strategy), clean text fields, parse the timestamp, impute missing pollutant values, classify readings into AQI-style categories, aggregate by geography and pollutant, and present results through interactive Streamlit + Plotly components. Streamlit handles the application layer and state; Pandas performs the tabular processing; NumPy backs numeric operations; Plotly Express renders the interactive charts and map.

9. Exploratory Data Analysis & Findings

The following are descriptive statistics computed directly from the cleaned dataset. They characterise this snapshot and dataset only, not a comprehensive assessment of India's air quality.

Pollutant	Obs.	Mean	Median	Min	Max
PM10	502	76.86	63.50	12.00	342.00
PM2.5	502	68.26	49.50	2.00	405.00
CO	502	37.26	31.00	2.00	195.00
OZONE	502	31.50	26.00	0.00	268.00
NO2	502	23.51	21.00	0.00	163.00
SO2	502	14.58	12.00	0.00	108.00
NH3	502	8.53	5.00	0.00	31.00

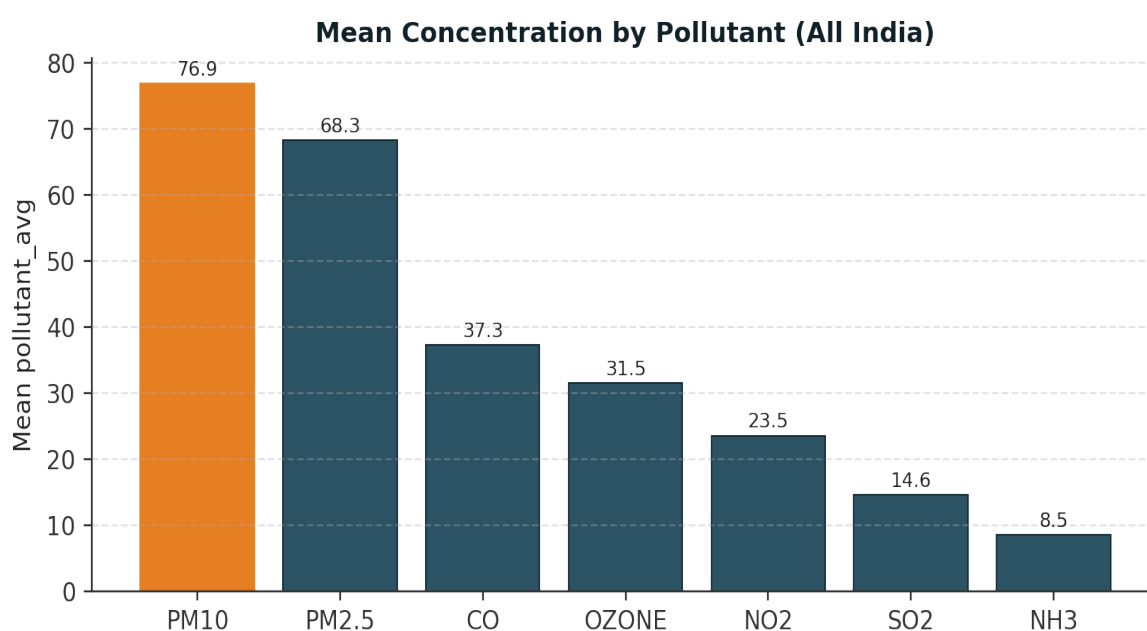


Figure 1 — Mean pollutant_avg across all stations, by pollutant type.

Across all pollutant observations combined, the state-level mean of pollutant_avg is highest for **Ladakh (81.14)** and lowest for **Sikkim (6.14)**. Because different pollutants are measured on different scales, this combined figure should not be read as a universal air-quality ranking.

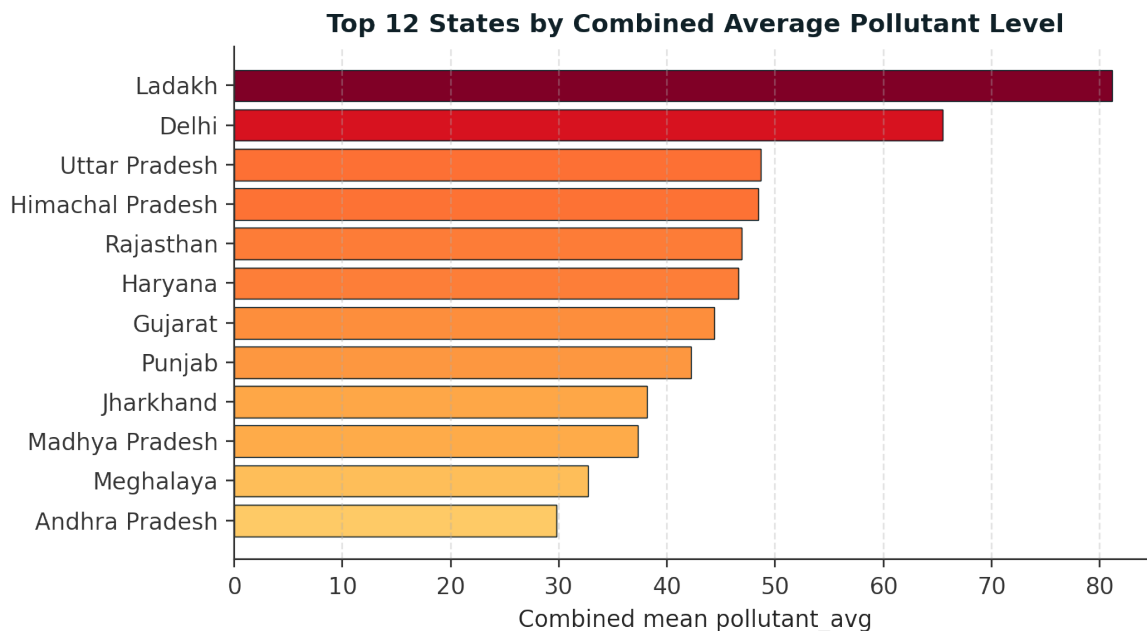


Figure 2 — Top 12 states by combined mean pollutant_avg across all pollutant types.

At city level, the highest combined mean pollutant_avg is observed for **Khora (101.86)**, while the lowest is **Gangtok (6.14)**. Using the CPCB PM2.5 breakpoints specifically, of the 502 PM2.5 station readings in this snapshot, 137 fall in the Good band, 173 Satisfactory, 78 Moderate, 30 Poor, 82 Very Poor and 2 Severe.

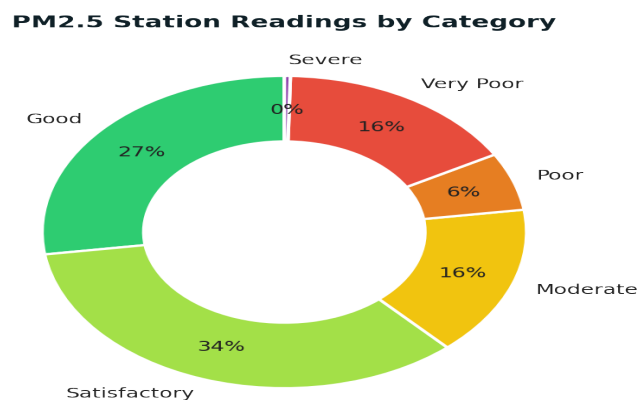


Figure 3 — Share of PM2.5 station readings by CPCB-style category, this snapshot.

10. Technical Implementation

The Streamlit application is contained in a single `app.py` file. Data loading and cleaning are cached with Streamlit's `@st.cache_data` mechanism so repeated filter interactions do not re-read or re-clean the CSV. Global state, city and pollutant filters are resolved before any analytical tab renders; each tab then filters the already-cleaned DataFrame in place.

- Plotly renders the station ranking bars, category donut, geographic map, grouped bar charts and box plots — all filter-responsive and hoverable.

- The map switches between a continuous colour scale (raw concentration) and a discrete, CPCB-coloured scale (category) via a radio control.
- A CSV export of the currently filtered data is available directly from the Comparison tab.

11. Limitations

- The dataset is a single real-time snapshot; it does not support genuine trend-over-time analysis, only a cross-sectional view of one moment.
- Pollutant values may use different measurement units, so cross-pollutant numeric comparisons require care.
- Median imputation for missing pollutant measures may understate the true variability of those fields.
- Non-PM pollutant categories (CO, NO₂, SO₂, OZONE, NH₃) use dataset-relative quartile labels rather than official CPCB AQI sub-index calculations, since exact reporting units are not confirmed.
- Correlation, causal inference and health-risk assessment remain outside the scope of this dashboard.

12. Future Scope

- A formal, always-visible data-quality panel (missing %, duplicate rows, parsing failures).
- Pollutant-specific unit confirmation and full CPCB AQI sub-index calculation for all seven pollutants.
- Historical ingestion so genuine time-series trend charts become meaningful.
- Station-level drill-down pages and anomaly detection for outlier readings.
- Validated forecasting models, clearly separated from the current descriptive-only insights.

13. Conclusion

AIRWISE provides a portfolio-ready framework for exploring India's air-quality observations through an interactive Streamlit interface. This revision closes three concrete defects — a file loading mismatch, a chart that assumed time-series data the dataset does not contain, and an incorrect KPI condition — and replaces a cluttered default filter view with a scoped, search-friendly filter system and a cleaner, tabbed layout. The result keeps every generated insight traceable to the underlying dataset while making the interface easier to navigate.